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**Agentic AI Research & Patent Gap Discovery
Framework**

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Table of Contents

Page No

Project Title	3
Team / Group Formation:	3
Introduction.....	3
Problem Statement	3
Project Objectives.....	4
Project Scope.....	4
Literature Survey	5
Project Implementation Methodology	11
Technologies & Tools to be used	14
References	15

Project Title

Agentic AI Research & Patent Gap Discovery Framework

Team / Group Formation:

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Introduction

Artificial Intelligence is increasingly being used to support research, innovation, and analysis of large amounts of technical information. In particular, **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, **semantic search**, and **Multi-Agent AI systems** have made it possible to automatically understand and analyze large collections of documents.

When researchers or innovators develop a new idea, one of the first questions is whether a similar idea has **already been patented or published**. Finding this information usually requires searching patent databases and research papers, reading lengthy technical documents, and manually comparing the proposed idea with existing work. This process can be difficult and time-consuming.

The proposed project, **Agentic AI Research & Patent Gap Discovery Framework**, aims to develop an AI-based system that helps automate this initial analysis. The system will retrieve relevant patents from **Google Patents** and research papers from **arXiv**, analyze their similarity with a given idea, and use multiple AI agents to assess novelty and identify potential gaps in existing research and patents.

The project will also compare three approaches : **Single LLM**, **LLM with RAG**, and **Multi-Agent AI** to determine which approach provides better and more explainable results for early-stage invention assessment.

The main goal is therefore not just to build an AI application, but to investigate whether **agent-based collaboration can improve the accuracy and explainability of patent and research-gap discovery compared with simpler LLM-based approaches**.

Problem Statement

Researchers and innovators often have to manually search through patents and research papers to determine whether a new idea already exists or whether there is a potential gap for further innovation. This process is time-consuming and difficult due to the large volume and complexity of available information.

Therefore, the project aims to develop an AI-based framework for early-stage patent and research-gap discovery that will take an invention or research idea as input, automatically

retrieves related information from relevant patents and research papers(**Google Patents and arXiv**), evaluates the novelty of a proposed idea, and identifies potential research and patent gaps and provide an explainable assessment of the idea using **LLMs, RAG, semantic similarity, and Multi-Agent collaboration**.

Project Objectives

The proposed project aims to combine these capabilities into a single framework :-

1. **Patent and research paper retrieval** is required to provide relevant prior-art information from both **Google Patents and arXiv** instead of depending on a single source.
2. **RAG** is included to allow the LLM to base its analysis on retrieved documents rather than relying only on its pre-trained knowledge.
3. **Multi-Agent AI** is used to divide the analysis into specialized tasks such as retrieval, document analysis, novelty assessment, and gap identification.
4. **Novelty scoring and explainable reasoning** are included so that the system does not simply label an idea as novel or non-novel, but also explains the evidence behind its assessment.
5. **Comparison of Single LLM, LLM + RAG, and Multi-Agent approaches** is necessary to experimentally determine whether the additional complexity of RAG and agent collaboration actually improves the quality of novelty assessment.
6. **Research and patent-gap discovery** is the final objective, allowing the system to identify areas where existing patents and publications provide limited coverage and where further innovation may be possible.

Thus, the objectives are designed to address the major limitations found in existing studies while keeping the project feasible within the scope of a semester-long PBL project.

Project Scope

The main Scope of the project is :-

- Retrieve relevant patents and research papers related to a given idea.
- Identify similar technologies, methods, and existing work.
- Estimate the novelty of the proposed idea.
- Identify potential research and patent gaps.
- Provide an explainable report showing why the idea is considered similar or different from existing work.
- Compare **Single LLM, LLM + RAG, and Multi-Agent** approaches.

The project is expected to produce :-

- A working prototype of the proposed framework.
- A small, manually verified evaluation dataset of invention ideas and their known prior art.
- Comparative experimental results for the three approaches.

- Novelty/gap assessment reports with supporting evidence.
- Evaluation results using suitable metrics such as **Precision, Recall, F1-score, explainability, and response time**.
- A research paper based on the experimental findings.

To keep the project feasible within the semester, the scope will be limited to:

- **Domain** : AI/ML-related inventions and research.
- **Data Sources** : Google Patents and arXiv.
- **Analysis** : Primarily text-based patent and research-paper analysis.
- **System Type** : Research prototype, not a replacement for professional patent examination or legal advice.
- **Evaluation** : A defined set of approximately **20–30 manually verified test cases**.
- **Comparison** : Single LLM, LLM + RAG, and Multi-Agent framework.

Full-scale USPTO/EPO database integration, multilingual analysis, legal patent filing, and complete commercial patentability assessment are **outside the scope of the current project** and may be considered as future work.

Literature Survey

Table 1 : Literature Review of 20 Research Papers

Title & Authors (Year)	Dataset Used	Sample Size	Algorithm(s)	Method/Approach	Accuracy / F1 / AUC	Key Finding	Research Gap / Limitation
1. Patent Intelligence in the Age of AI – Giuntelli et al. [2026]	Orbit Intelligence, Robotics patents, Horizon Europe	2,726 patents; 800K+ corpus	Gemini LLM, Semantic Search, NLP	AI semantic classification + clustering + RAG workflow	Precision Recall Manual Validation	Enables scalable patent intelligence, technology mapping, trend analysis and white-space discovery using AI	Patent-only; no Multi-Agent; no novelty scoring; commercial APIs; limited quantitative evaluation

2. Agent Ideate – Kanumolu et al. [2025]	AgentScen dataset, DuckDuckGo	150 patents	Llama-4, CrewAI	Patent → Keyword Extraction → Web Search → Idea Generation → Validation	Human Evaluation; LLM Judge	Multi-agent workflow produces higher-quality business ideas than single LLMs	No novelty assessment; no RAG; no semantic retrieval; no patentability evaluation
3. VIRSCI – Dong et al. [2025]	Open Academic Graph, Semantic Scholar	Large-scale publication database	GPT-4o, FAISS, Multi-Agent	Collaborative virtual scientific ecosystem with RAG	Novelty; Human Review; Feasibility	Produces higher-quality and more novel scientific ideas through agent collaboration	Scientific papers only; no patents or patentability assessment
4. EvoPat – Wang et al. [2024]	Google Patents, Semantic Scholar	5,000 patents	GPT-4o, AutoGen, FAISS, BGE-M3	Five specialized patent-analysis agents + RAG	ROUGE; BERTScore; Human Evaluation	Generates comprehensive patent summaries with collaborative reasoning	Focuses on patent summarization, not invention novelty assessment
5. Can AI Examine Novelty of Patents? – Ikoma et al. [2025]	USPTO, Google Patents	3,975 patent applications	GPT-4o, Llama-3, Longformer	Claim vs Prior-Art comparison	Accuracy; Precision; Recall; F1	First benchmark for AI-based patent novelty evaluation	No RAG; no Multi-Agent; prior art assumed available
6. PatentAgent (TCL-MFSNE) – Authors et al. [2025]	PRAO Patent Dataset	8,579 training pairs	GPT-4o-mini, Qwen, Contrastive Learning	Multi-Agent + Byzantine Consensus	Accuracy; Precision; Recall; F1	~85% accuracy with improved robustness and reduced hallucinations	No RAG; patent-only; computationally expensive

7. Exploring Design of Multi-Agent LLM Dialogues for Research Ideation – Ueda et al. [2025]	Semantic Scholar	7,000 generated ideas	GPT-4o-mini	Ideation → Critique → Revision workflow	Precision @k; GPT Ranking	Multi-agent dialogue significantly improves research idea quality	Not designed for patent analysis
8. Automatic Patent Literature Retrieval using LLM-RAG – Ding et al. [2025]	Google Patents	Millions of patents	GPT-4, FAISS	Dense Retrieval + RAG	Accuracy ; Recall	Achieved 80.5% accuracy and 92.1% recall for patent retrieval	Retrieval only; no novelty assessment
9. Evaluation Methods for Patent Novelty Search Systems – Zhang et al. [2025]	Patent citation datasets	10,000 patents	PatSnap, RL similarity model	Citation-based novelty search evaluation	Recall; Top-k Detection	Standardized evaluation framework for patent search systems	No LLM; no reasoning; evaluation-only
10. Tree-of-Claims (ToC) – Yu et al. [2026]	USPTO Office Actions	1,145 patents	GPT-4o, Claude, MCTS	Examiner Agent + Editor Agent + MCTS	ROUGE; BLEU; Expert Preference	Improved patent claim optimization by ~8–9%	No retrieval; no novelty assessment

11. IdeaForge – Bose [2026]	Knowledge Graph, arXiv	5 application domains	TinyLlama, Knowledge Graph	Multi-Agent + Knowledge Graph	InnovationScore	Generates explainable patent drafts using multiple innovation methodologies	Uses only arXiv; heuristic scoring; no Google Patents
12. AgentSwift – Li et al. [2026]	Seven benchmark tasks	220 labeled samples	MCTS, Bayesian Optimization	Hierarchical agent optimization	Success Rate; R^2	Automatically discovers high-performing LLM agent architectures	General-purpose; not patent-specific
13. TCLMA – Zheng et al. [2025]	PRAO Patent Dataset	8,579 training pairs	GPT-4o-mini, Qwen, FAISS	Multi-Agent + RAG + Contrastive Learning	Accuracy ; Precision ; Recall; F1	85% accuracy , better than Single LLM and Implicit LLM	Patent-only; no scientific literature integration
14. IPAS-CARNA – Han et al. [2026]	CLEF-IP, USPTO, HUPD	100,000 claim pairs	Longformer, BigBird	Claim-aware retrieval + continuous novelty scoring	MAP; NDCG; Accuracy ; F1	Integrates retrieval, novelty assessment and summarization	No LLM; no Multi-Agent; patent-only
15. VIRSCI – Kayakökü et al. [2025]	Open Academic Graph	Millions of papers	Llama-3, AgentScope	Multi-Agent + RAG	Novelty; Feasibility	Medium-sized agent teams generated the highest novelty ideas	Scientific literature only; no patent analysis
16. AgentSwift – Li et al. [2026]	ALFWorld, SciWorld , WebShop	7 benchmarks	GPT-4o, DeepSeek, MCTS	Value-guided hierarchical search	Success Rate, R^2	Improved automated LLM agent design by 8.34%	No patent retrieval or novelty evaluation

17. MemGraph – Xiong <i>et al.</i> [2025]	PatentMatch, Patent Corpus	300,000 patents	Llama-3, Qwen, Memory Graph	Memory Graph + RAG	Accuracy, IPC Accuracy	Improved patent matching by 17.68% over vanilla LLMs	No Multi-Agent, no novelty assessment
18. AI-based Novelty Detection in Crowdsourced Idea Spaces – Just <i>et al.</i> [2024]	Crowdsourced ideas	232 ideas	SBERT, GPT-3 Embeddings	Embedding similarity + kNN	Spearman Correlation	SBERT best matched human novelty judgments	Small dataset; no patents; no explainability
19. Technology Knowledge Elements Framework – Wei <i>et al.</i> [2024]	USPTO Agricultural Robot Patents	1,343 patents	BERT, UIE, HGT	Knowledge Graph + Triple Novelty	Accuracy, F1	Triple-level novelty evaluation with 72% accuracy	No LLM, no RAG, domain-specific
20. RAG Systems for Intellectual Property – Ren <i>et al.</i> [2025]	Taiwan Patent Q&A	~50,000 queries	GPT-4, DeepSeek, PatentSBERTa	Synthetic Query Generation + RAG	MRR, Hit@k, ROUGE, BLEU	Improved retrieval by 185–262% over baselines	No Multi-Agent, no novelty assessment
21. Agentic AI Research and Patent Gap Discovery Framework – Alvira Parveen, Rohit [2026]	Google Patents + arXiv; AI/ML-related patent and research-paper corpus	20-30 verified test cases	LLMs, Sentence-Transformer embeddings, LangChain/LangGraph; RAG	Multi-Agent Retrieval + Prior-Art Analysis + Novelty Scoring + Gap Discovery	Precision, Recall, F1-score; novelty-score correlation with human evaluation.	End-to-end patent & research-gap discovery with explainable novelty assessment; compares Single LLM vs LLM+RAG vs Multi-Agent	Limited to AI/ML domain; hallucination/retrieval errors; computational cost; novelty requires human validation

Table 2 : Summary of the 20 Research Papers

Focus/Approach	Example Studies	Key Finding/Result	Limitation
1. Patent Intelligence & Semantic Patent Analysis ; uses LLMs, semantic representations, clustering and RAG to understand large patent collections and identify technologies, competitors and white spaces.	Giuntelli et al. (2026), <i>Patent Intelligence in the Age of AI</i>	AI-based semantic representations improve patent retrieval/classification compared with conventional keyword approaches and can generate technology–application matrices, competitive maps and white-space insights.	Mainly focuses on patent analytics and classification , not on automatically generating and validating new research ideas. Large patent documents also create processing and grounding challenges.
2. Patent Retrieval & Prior-Art Search ; uses embeddings, vector search, LLM-RAG and claim-document matching to retrieve relevant patents/prior art.	Ding et al. (2025), <i>Automatic Patent Literature Retrieval using LLM-RAG</i> ; Han & Qu (2026), <i>IPAS-CARNA</i> ; Zhang et al. (2025)	LLM-RAG substantially improves semantic retrieval. Ding et al. reported 80.5% semantic matching accuracy and 92.1% recall , while IPAS-CARNA integrates retrieval, novelty scoring and summarization.	Retrieval systems may identify similar documents without explaining which claim elements create the actual gap . Fixed chunks, maximum-similarity scoring and incomplete claim-element coverage remain problems.
3. Patent Novelty / Patentability Detection ; compares patent claims with prior-art documents using LLMs, Siamese networks, contrastive learning and classification models.	Ikoma & Mitamura (2025); Zheng et al. (2025) TCLMA; Kayakökü & Tüfekci (2025)	AI can assist novelty assessment. TCLMA reports approximately 85% accuracy and F1-score , while generative LLMs showed reasonable ability to assess novelty and explain claim–prior-art relationships.	Most approaches mainly answer “ novel or not novel? ” rather than discovering the underlying research/technology gap and generating a validated new direction. Classification models can also struggle with complex claim–prior-art correspondence.
4. Multi-Agent Patent Analysis & Summarization ; assigns different LLM agents to tasks such as summarization, innovation identification, comparison, search and evaluation.	Wang et al. (2024), <i>EvoPat</i>	Multi-LLM collaboration and RAG provide richer patent summaries, innovation analysis, technical comparisons and strengths/weaknesses than a single LLM. The system also uses Google Patents and Semantic Scholar tools to obtain external information.	Primarily provides analysis and summarization . It does not completely connect patent analysis with novelty-gap discovery and autonomous research-idea generation.
5. Patent-to-Product / Innovation Idea Generation ; uses multi-agent LLMs to read patent information and generate product concepts, implementation ideas and differentiation.	Kanumolu et al. (2025), <i>Agent Ideate</i>	Tested on 150 U.S. patents from Computer Science, NLP and Material Chemistry. Agentic approaches outperformed standalone LLMs in idea quality, relevance and novelty.	Generated ideas are evaluated mainly for quality, feasibility, innovation and market relevance ; they are not necessarily checked against comprehensive patent prior art. Therefore, an idea that appears novel to an LLM may already exist.
6. Multi-Agent Scientific Ideation & Idea Refinement ; multiple LLM agents generate, critique, discuss and refine research ideas through iterative collaboration.	Su et al. (2025), <i>VIRSCI</i> ; Ueda et al. (2025), <i>Multi-Agent LLM Dialogues for Research Ideation</i>	Multi-agent collaboration improves idea diversity and novelty. Ueda et al. found that increasing agent diversity and interaction depth improves ideation; around 2–3 refinement turns provided strong results.	These systems focus on scientific idea generation , not patent-level novelty. Evaluation may rely heavily on LLM judges; Ueda et al. explicitly identify automatic GPT-4 evaluation as a limitation.

7. Knowledge-Graph / Graph-RAG Based Innovation Reasoning ; uses knowledge graphs to preserve relationships between technologies, claims, ideas and prior knowledge while enabling multi-agent reasoning.	Zhou et al. (2025), <i>Graph-based RAG</i> ; Bose (2026), IdeaForge	Graph-based RAG can improve retrieval and contextual reasoning. IdeaForge combines TRIZ + Design Thinking + SCAMPER through a persistent knowledge graph and identifies convergence between independently generated claims.	Convergence does not guarantee legal novelty. IdeaForge's prior-art search is incomplete because it uses arXiv rather than comprehensive patent databases; semantic similarity can also miss or falsely identify equivalent ideas.
8. Multi-Agent Patent Claim Optimization ; uses specialized agents and search/planning techniques to iteratively modify patent claims while balancing novelty, legal scope and semantic consistency.	Yu et al. (2026), Tree-of-Claims (ToC)	MCTS combined with EditorAgent and ExaminerAgent improved patent-claim revision performance, with an average 8% composite-score improvement on 1,145 claims.	Focuses on improving existing claims , rather than discovering completely new research/technology gaps. It also depends on simulated examiner reasoning rather than actual legal examination.
9. AI-Based Novelty Measurement of Ideas ; uses embeddings such as SBERT/Doc2Vec to compare ideas against existing solutions and estimate novelty.	Just et al. (2024), <i>AI-based Novelty Detection in Crowdsourced Idea Spaces</i>	Embedding-based approaches can approximate human novelty judgments, particularly when a suitable reference set of existing solutions is available.	Novelty scores depend strongly on the reference dataset, text length and writing style . A semantically unusual description can receive a high score even when the underlying idea is not actually new.
10. Automated Agent Architecture / Agent Search ; automatically searches for effective combinations of workflows, planning, memory and tools instead of manually designing an agent.	Li et al. (2026), AgentSwift	AgentSwift combines hierarchical search, a value model and MCTS and reports an average 8.34% performance improvement across seven benchmarks.	It is a general-purpose agent-design framework , not specifically designed for patent intelligence. Searching and evaluating many candidate agent architectures can also be computationally expensive.

Project Implementation Methodology

Step 1) Define the Invention Idea

The user provides a brief description of an **AI/ML-related invention or research idea**, including its main concept, technical features, and intended application. This idea becomes the input for the complete analysis.

Step 2) Data Collection

Relevant information is collected from the two selected sources :-

- **Google Patents** : for existing patents and prior-art information.
- **arXiv** : for related research papers and scientific work.

A defined set of relevant documents will be collected and prepared for experimentation.

Step 3) Data Preprocessing

The retrieved patents and research papers are cleaned and prepared for analysis. Important information such as **title, abstract, claims, technical description, and publication details** is extracted where available. Unnecessary text and duplicate records are removed.

Step 4) Semantic Representation

The invention idea and retrieved documents are converted into numerical **embeddings** using a Sentence Transformer or similar embedding model. This allows the system to compare the meaning of technical concepts rather than relying only on exact keywords.

Step 5) Semantic Retrieval

Cosine similarity is used to compare the invention idea with the available patents and research papers. The most relevant documents are ranked and selected as potential prior art or related research.

Step 6) Retrieval-Augmented Generation (RAG)

The retrieved documents are provided as context to the LLM through a **RAG pipeline**. This allows the model to base its analysis on retrieved evidence instead of relying only on its pre-trained knowledge.

Step 7) Multi-Agent Analysis

The proposed Multi-Agent framework divides the analysis into specialized tasks:

- **Retrieval Agent** – identifies relevant patents and papers.
- **Patent Analysis Agent** – examines patent claims and technical features.
- **Research Analysis Agent** – examines related research papers and their contributions.
- **Novelty Agent** – compares the proposed idea with the retrieved prior art.
- **Gap Discovery Agent** – identifies areas with limited existing coverage.
- **Report Agent** – combines the findings into a final structured report.

Step 8) Novelty Assessment

The system evaluates the proposed idea using factors such as **semantic similarity, technical overlap, claim similarity, and retrieved evidence**. These factors are combined to generate a preliminary novelty assessment and score.

Step 9) Patent and Research Gap Identification

The system analyzes the differences between the proposed idea and existing work to identify:

- Existing technologies that are already well covered.
- Similar patents and publications.
- Features that appear less explored.
- Potential research or innovation opportunities.

Step 10) Explainable Result Generation

Instead of providing only a numerical score, the system generates an explanation describing **why the idea appears similar or different from existing work** and identifies the patents or papers supporting the assessment.

Step 11) Comparative Evaluation

The same set of test cases will be evaluated using three approaches:

1. **Single LLM**
2. **LLM + RAG**
3. **Proposed Multi-Agent + RAG framework**

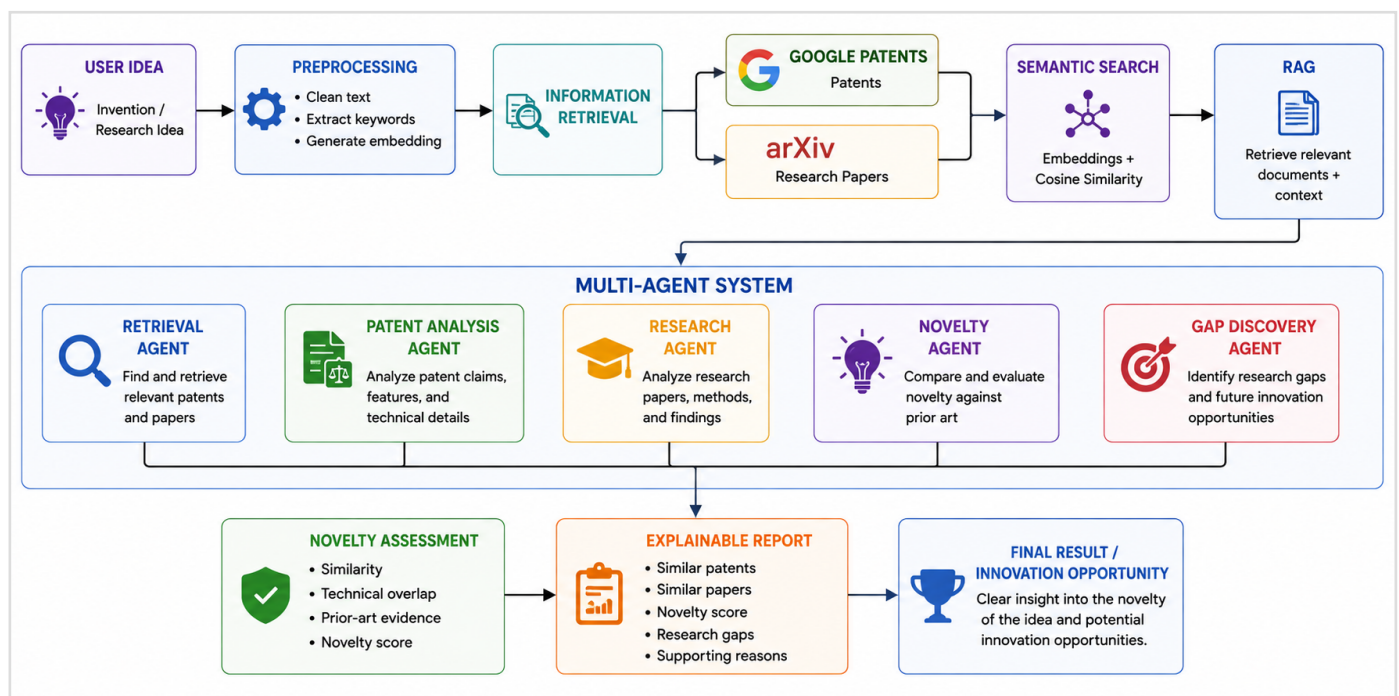
Their results will be compared using **Precision, Recall, F1-score, explainability, and response time**, along with human-verified ground-truth assessments.

Step 12) Final Analysis and Validation

The results from all three approaches will be analyzed to determine whether the proposed Multi-Agent framework provides better and more reliable novelty and gap assessments than the simpler approaches.

The final outcome will be a **working prototype, comparative experimental results, and an explainable patent/research-gap assessment report**, which will form the basis of the proposed research paper.

System Architecture



Technologies & Tools to be used

Table 3 : Hardware Requirements

Component	Minimum Requirement	Recommended
Processor	Intel Core i5 / AMD Ryzen 5	Intel Core i7 / AMD Ryzen 7
RAM	8 GB	16 GB
Storage	256 GB SSD	512 GB SSD
GPU	Not mandatory	NVIDIA GPU with 4 GB+ VRAM, if local models are used
Internet	Stable internet connection	High-speed internet connection

Table 4 : Technology Stack

Technology	Role in the Project
Large Language Model (LLM)	Understanding technical documents, reasoning and generating explanations
LangChain / LangGraph	Building and coordinating the Multi-Agent workflow
Retrieval-Augmented Generation (RAG)	Providing retrieved patent/paper information as context to the LLM
Sentence Transformers	Creating semantic embeddings of ideas, patents and research papers
FAISS / ChromaDB	Storing and retrieving document embeddings
Cosine Similarity	Measuring semantic similarity between the proposed idea and existing work
Hugging Face Transformers	NLP and embedding-model support
Scikit-learn	Similarity calculations and evaluation metrics
Pandas / NumPy	Data preprocessing and analysis
Matplotlib	Visualization of experimental results

Table 5 : Software Requirements

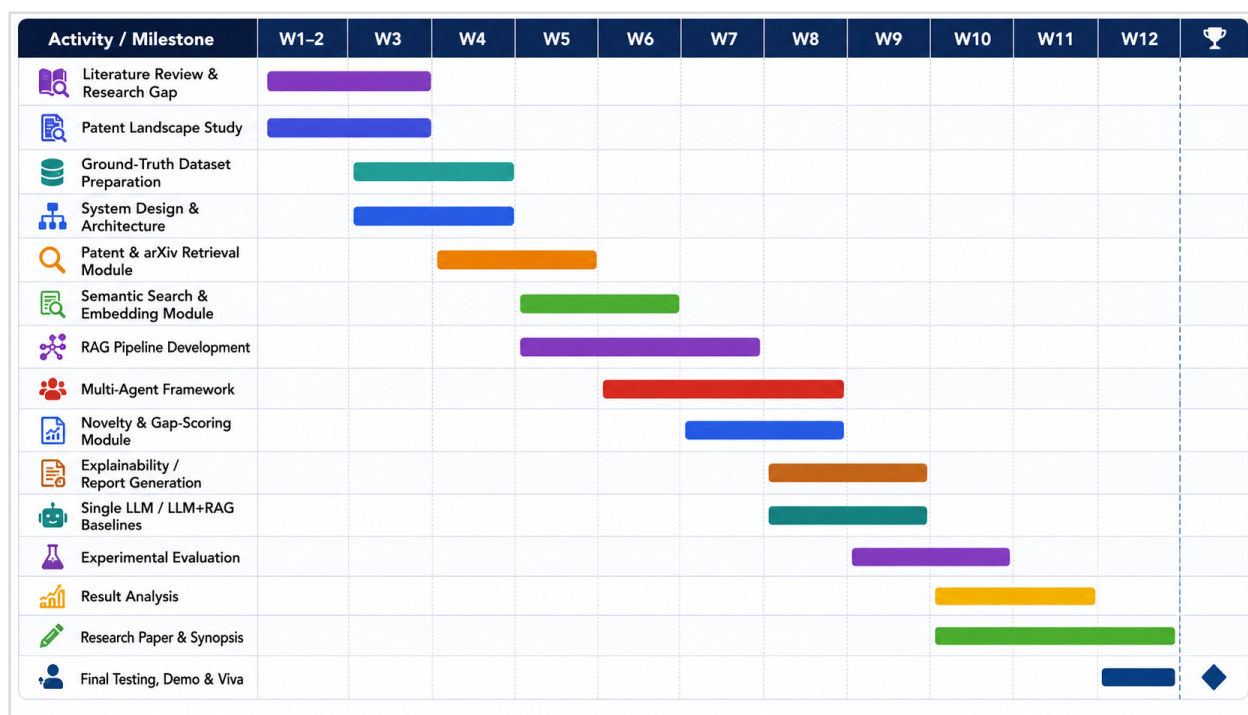
Software / Tool	Purpose
Windows 10/11 / Linux	Operating system
Python 3.11+	Main programming language
VS Code / PyCharm	Development environment
Jupyter Notebook	Experiments and data analysis

Git & GitHub	Version control and team collaboration
Web Browser	Accessing APIs, patent/research sources and testing

Table 6 : Data Sources / APIs

Source	Purpose
Google Patents	Retrieval and analysis of existing patents/prior art
arXiv	Retrieval of related scientific research papers
LLM API	Reasoning, document analysis and explainable report generation

Project Timeline



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Signature

Student Name	Student Sign	Faculty Name	Faculty Sign